

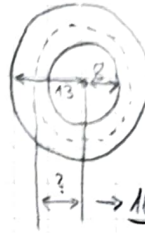
(6) $\rightarrow$ Permeabilität $\mu = ?$

$L = \frac{\mu N^2 A}{l}$ (4.104) mit $l = 200 \mu\text{H}$, $N = 36$ Wdg.,

$$200 \mu\text{H} = \frac{\mu \cdot 36^2 \cdot 60 \text{ mm}^2}{21 \pi \text{ mm}}$$

$$l \cdot l / (\mu_0 \cdot N^2 \cdot A) = \mu_r$$

$$\mu_r = \underline{135}$$

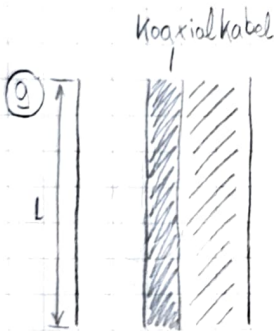


$$\rightarrow 10,5 \text{ mm} = r$$

$$U = 21 \pi = l$$

$$A = a \cdot b \quad \text{mit } b = 12 \text{ mm}$$

$$A = (13 - 8) \cdot 12 \text{ mm} = 60 \text{ mm}^2$$



$$\Phi = \int_{R_1}^{R_2} B(r) dA(r) \quad \text{(Folie 196)}$$

$$\Phi = \iint_A \vec{B} dA \quad \text{(Folie 151)}$$

$$l = \frac{\Phi}{I}$$

$$\oint \vec{H}(r) ds = I$$

$$\vec{H}(r) \cdot \underbrace{2\pi r}_{\text{Umfang}} = I$$

$$\vec{H}(r) = \underline{\underline{\frac{I}{2\pi r}}}$$

$$\vec{B}(r) = \mu \cdot \vec{H}(r)$$

$$\vec{B}(r) = \underline{\underline{\frac{\mu I}{2\pi r}}}$$

$$\Phi = \iint_A \vec{B}(r) \cdot d\vec{A}$$

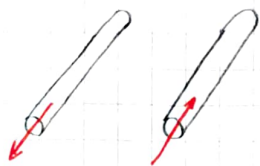
$$\Phi = \int_0^l \int_{r_i}^{r_o} \frac{\mu \cdot I}{2\pi r} \cdot dr \cdot dz$$

$$\Phi = \int_0^l \frac{\mu \cdot I}{2\pi} \ln\left(\frac{r_o}{r_i}\right) \cdot dz$$

$$\Phi = \frac{\mu \cdot I \cdot l}{2\pi} \cdot \ln\left(\frac{r_o}{r_i}\right) \rightarrow \text{einsetzen in } l = \frac{\Phi}{I}$$

$$l = \frac{\mu \cdot I \cdot l}{2\pi} \cdot \ln\left(\frac{r_o}{r_i}\right) = \underline{\underline{\frac{\mu \cdot l}{2\pi} \cdot \ln\left(\frac{r_o}{r_i}\right)}}$$

(10)



$$\Phi = \vec{B}(r) \cdot d\vec{A}$$

$$\oint \vec{H}(r) ds = I$$

$$\vec{H}(r) \cdot 2\pi r = I$$

$$H(r) = \frac{I}{2\pi r}$$

$$B(r) = \frac{\mu \cdot I}{2\pi r}$$

$$\Phi = \Phi_1 + \Phi_2 = 2\Phi = 2 \int_0^l \int_{rd}^{a-rd} \frac{\mu \cdot I}{2\pi r} dr dz = \frac{\mu I l}{2\pi} \cdot \ln\left(\frac{a-rd}{rd}\right)$$

$rd \ll a \Rightarrow rd$ kann vernachlässigt werden!

$$L = \frac{\Phi}{I} = \frac{\frac{\mu I l}{2\pi} \cdot \ln\left(\frac{a}{rd}\right)}{I} = \frac{\mu l}{\pi} \cdot \ln\left(\frac{a}{rd}\right)$$

(13)



$\rightarrow$ abstoßende Kraft zwischen den Leitern

$$W = \frac{1}{2} L \cdot I^2$$

$$F = \frac{dW}{ds} = \frac{d}{da} \frac{\mu l \cdot \ln\left(\frac{a}{rd}\right)}{2\pi} I^2$$

$$F = \frac{\mu l \cdot I^2}{2\pi a}$$

bei $I=1A$; $a=1m$

$$\frac{F}{l} = \frac{4\pi \cdot 10^{-7} \frac{Vs}{Am} \cdot 1 A^2}{2\pi \cdot 1m}$$

$$\frac{F}{l} = 2 \cdot 10^{-7} \frac{J}{m^2} \Rightarrow \frac{N}{m}$$